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REVIEW ARTICLE

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Analgesic and anti-inflammatory drugs in sports: Implications for exercise performance and training adaptations

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Over-the-counter analgesics, such as anti-inflammatory drugs (NSAIDs) and paracetamol, are widely consumed by athletes worldwide to increase pain tolerance, or dampen pain and reduce inflammation from injuries. Given that these drugs also can modulate tissue protein turnover, it is important to scrutinize the implications of acute and chronic use of these drugs in relation to exercise performance and the development of long-term training adaptations. In this review, we aim to provide an overview of the studies investigating the effects of analgesic drugs on exercise performance and training adaptations relevant for athletic development. There is emerging evidence that paracetamol might acutely improve important endurance parameters as well as aspects of neuromuscular performance, possibly through increased pain tolerance. Both NSAIDs and paracetamol have been demonstrated to inhibit cyclooxygenase (COX) activity, which might explain the reduced anabolic response to acute exercise bouts. Consistent with this, NSAIDs have been reported to interfere with muscle hypertrophy and strength gains in response to chronic resistance training in young individuals. Although it remains to be established whether any of these observations also translate into detriments in sport-specific performance or reduced training adaptations in elite athletes, the extensive use of these drugs certainly raises practical, ethical, and important safety concerns that need to be addressed. Overall, we encourage greater awareness among athletes, coaches, and support staff on the potential adverse effects of these drugs. A risk-benefit analysis and professional guidance are strongly advised before the athlete considers analgesic medicine for training or competition.

KEYWORDS

acetaminophen, endurance, ibuprofen, muscle adaptations, nonsteroidal anti-inflammatory drugs, paracetamol, recovery, strength

1 | INTRODUCTION

Analgesic drugs are medicines that are used to relieve pain (ie, "painkillers") without loss of consciousness or sensory perception (as opposed to anaesthetics). A large number of medicines have these properties, including opioids (narcotics), nonsteroidal anti-inflammatory drugs (NSAIDs), and paracetamol. While opioids require prescription and are banned within sports, NSAIDs (such as ibuprofen, indometacin, ketoprofen, naproxen, acetylsalicylic acid, and diclofenac) and paracetamol are sold over-the-counter and are currently not classified as doping agents. The use of over-the-counter analgesic drugs is commonplace in elite sports as well as in recreational and student-athletes.[1,2] It has been reported that the use of NSAIDs is much higher in Olympic athletes compared with age-matched controls.[1,3] In addition,

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both male and female football players participating in the FIFA World Cup, over 50% of the players used NSAIDs at least once during the tournament, and about 7 players per national team were using NSAIDs prior to every match.[4] These authors highlighted that the reported frequency of NSAID use is probably underestimated, since self-medication or treatment already prescribed by club physicians is typically not included in published reports. The use of other analgesics such as paracetamol was far less common. This could be due to the fact that paracetamol, typically, has weaker anti-inflammatory effects than NSAIDs.[5] High use of NSAID medication has also been reported in collegiate athletes,[2,6] athletes at the Sydney Olympics in 2000,[7] in high-caliber track and field athletes,[8] triathletes,[9] and marathon runners.[10]

Athletes reported that they used NSAIDs to reduce pain and inflammation associated with training, competition or soft tissue injuries, or to gain a competitive advantage.[1,2] The high use of analgesic drugs takes place despite the fact that most sports organizations declare that unnecessary medication should be minimized due to potential short- and long-term adverse effects. For example, the FIFA Medical Assessment and Research Centre (F-MARC) campaigned to reduce the use of NSAIDs prior to the 2010 FIFA World Cup. Despite this initiative, the reported intake of NSAIDs remained unchanged from the 2010 to the 2014 tournament.[4]

One of the primary ways that over-the-counter analgesic drugs exert their pain-reducing effect is by inhibiting the activity of cyclooxygenase (COX), a family of enzymes that facilitate the production of prostaglandins (Figure 1). The COX enzymes exist in three isoforms (COX-1, COX-2, and a splice variant termed COX-3), and NSAIDs are often classified based on their specificity toward the two main isoforms. Nonselective COX inhibitors are generally considered to block both main isoforms, whereas specific inhibitors are more prone to block either COX-1 or COX-2.[11] Although paracetamol is not classified as an NSAID, it is now generally accepted that this drug has a spectrum of actions similar to NSAIDs and inhibits COX activity through the peroxidase function of these enzymes.[5] However, paracetamol also has antipyretic effects and central sites of actions (eg, in the central nervous system) that might differ from NSAIDs.[5]

Given the widespread use of analgesic drugs in the athletic population, a number of questions have been raised regarding the effects of these drugs on acute exercise performance and the influence on chronic training adaptations that are critical for long-term athletic development. It is particularly pertinent to study such effects given that the formation of prostaglandins may not only regulate pain and inflammation, but also modulate the protein turnover machinery controlling tissue remodeling for the adaptive responses to exercise. Thus, it makes the expectation tenable that if an athlete regularly uses analgesic drugs, to reduce pain associated with training or competition, there could be unwanted negative consequences for the long-term adaptive training response.

What is already known?

- Over-the-counter analgesics, such as anti-inflammatory drugs (NSAIDs) and paracetamol, are widely consumed by athletes worldwide.
- Although these drugs are used to reduce short-term pain and inflammation, they have also been shown to modulate muscle protein turnover through the effects on the cyclooxygenase (COX) enzyme pathways.

What is new?

- There is emerging evidence that ingestion of paracetamol might enhance important endurance parameters as well as aspects of neuromuscular performance, possibly through increased pain tolerance.
- Analgesic drugs have been shown to reduce the anabolic response to acute exercise bouts and attenuate long-term gains in muscle mass and strength in young healthy individuals.
- While further studies are required to better unravel the consequences of using analgesic drugs in elite athletes, the current data call for greater awareness among coaches, medicine, and science support staff with regard to potential adverse effects and the associated ethical issues surrounding the frequent use of these drugs.

Accordingly, the purpose of this article was to give an overview of the studies investigating the effects of analgesic drugs on exercise performance and training adaptations in contexts relevant for athletes. While NSAIDs are widely used by athletes to alleviate symptoms of muscle damage and delayed onset muscle soreness (DOMS), or to speed up the recovery of muscle function after such events, it is beyond the scope of this work and the reader is directed to previous reviews.[12,13] In addition, the influence of analgesic drugs for repair from a muscle injury will not be covered in detail in this article, and the reader is referred to other work.[14,15]

2 | EXERCISE PERFORMANCE

Given that the ability of an athlete to tolerate pain could be an important factor in high-intensity exercise performance, it is understandable that analgesic drugs have been investigated in the context of endurance and neuromuscular performance.

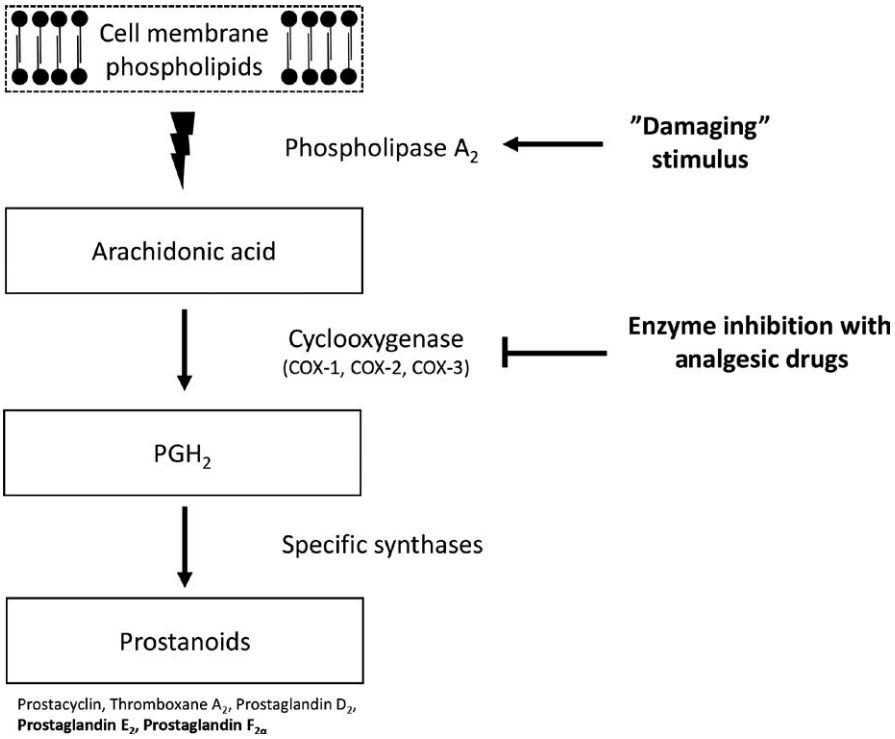

FIGURE 1 Schematic overview of prostaglandin biosynthesis in response to a damaging stimulus such as an intense exercise challenge or tissue injury. Prostaglandins are synthesized from arachidonic acid, which is liberated from the cell membrane. Analgesic drugs work in part by inhibiting cyclooxygenase (COX) activity, thereby reducing the production of several prostaglandins that have been implicated in skeletal muscle remodeling, particularly the ones highlighted in bold

There is indeed both anecdotal and research evidence of the use of narcotic analgesics in athletes long before the modern era of doping.[16] The nostalgic reader might recall the fate of one of Britain's most successful cyclists, Tom Simpson, who died on the slopes of Mont Ventoux during the 1967 Tour de France after he had taken amphetamine and alcohol, which contributed to a failure to effectively regulate exercise tolerance that led to a fatal case of heat exhaustion.

With regard to nonbanned substances, the first two studies conducted in the 1990s examined the effects of acetylsalicylic acid (aspirin). Lisse et al[17] gave 17 healthy male volunteers a single dose of 650 mg of aspirin or a placebo in a double-blind, crossover manner 30 minutes before running a 2 mile (3.2 km) time trial and reported no difference between the groups. A few years later, Roi et al[18] carried out a single-blind, crossover study to assess the effect of aspirin on maximal exercise capacity. In this work, 18 young men were given 1000 mg of aspirin, chewable buffered aspirin (1000 mg), or a placebo prior to completing a maximal incremental cycle ergometer test. The results showed no significant effect of the drug on ventilation, maximal oxygen uptake, heart rate, or blood lactate, supporting the earlier observation that a single dose of aspirin did not affect endurance performance.

More recently, Mauger et al[19] had 13 male trained cyclists to perform a self-paced 10-mile (16.1 km) cycle time trial, following either paracetamol (3 × 500 mg) or a placebo. In the paracetamol trial, completion time was faster and the participants cycled at a 4% higher mean power output with higher heart rate and blood lactate levels, but without changes in perceived pain or exertion. These findings suggested that exercise performance was, at least in part, regulated by pain perception, and hence, paracetamol could improve exercise performance through increased pain tolerance and lower perceived exertion. This is also supported by animal data showing that ibuprofen improved swimming performance (time to exhaustion) in rats, plausibly through reduced susceptibility to exercise-induced fatigue.[20]

In a follow-up study to the cycle time trial, the same group sought to determine whether paracetamol could also improve exercise capacity in the heat.[21] The background to this hypothesis comes from the fact that paracetamol has antipyretic effects for fever management, and perhaps also in afebrile patients.[22] The results of the study showed that an acute dose of paracetamol (20 mg/kg lean body mass) allowed participants (11 recreationally active men) to cycle longer in hot conditions (30°C), on average by a staggering 4 minutes (+17%). This was accompanied by significantly lower core, skin and body temperature, and lower perceived thermal strain.[21] The authors concluded that paracetamol may reduce the thermal challenge of exercise and therefore improve performance in hot conditions.

While the two aforementioned studies indicated improved endurance performance capacity with acute paracetamol consumption, Foster et al[23] explored the effect of paracetamol on repeated sprint-cycling performance. Nine recreationally active men completed 8 × 30 s sprints with 2 minutea active rest intervals in a randomized, crossover design after consuming either 1500 mg (3 × 500 mg) of paracetamol or a placebo. Following the ingestion of paracetamol, participants cycled at a greater mean power output during sprint 6, 7, and 8, and the relative decrement in mean power output was also reduced. The authors suggested that paracetamol improved performance through the reduction of self-perceived pain for a given work rate, thus enabling them to complete the exercise closer to their true performance limit. These results gave further credibility to the thesis that increased pain tolerance can improve exercise performance.

It has also been proposed that nociceptive afferent feedback and the associated sensation of pain might modulate both central and peripheral fatigue, hence playing a role in neuromuscular fatigue development.[24,25] To examine this idea, Morgan et al[26] had 13 active men to conduct 60 × 3 s maximum voluntary contractions separated by a 2 s passive recovery. The protocol was executed in a randomized crossover order after subjects ingesting either 1000 mg of paracetamol or a placebo. Mean torque was greater in the paracetamol trial compared with placebo, and this was associated with an attenuated decline in the EMG amplitude in the latter stages of the trial. Collectively, these findings suggested that paracetamol ingestion might improve repeated maximal

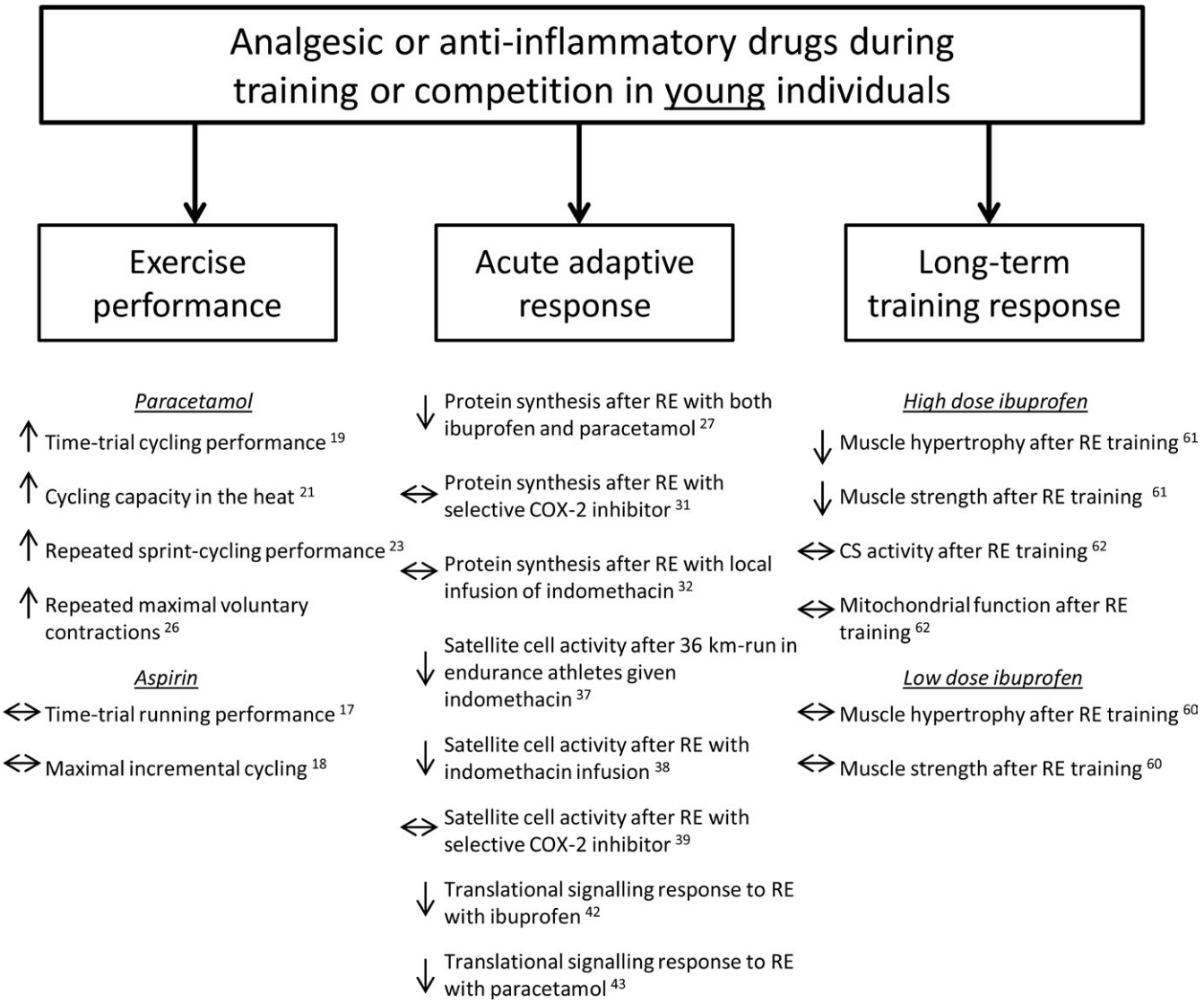

FIGURE 2 Summary of key studies reporting on the effects of analgesic or anti-inflammatory drugs on exercise performance or training adaptations in young individuals. RE = resistance exercise. Arrows pointing up indicate positive effect of the medical treatment, arrows pointing down indicate negative effect, and horizontal arrows indicate no effect

voluntary contractions by enabling a better preservation of muscle activation during exercise. As the attenuation of neuromuscular fatigue following paracetamol ingestion was not associated with altered maximal voluntary activation (central fatigue) or peripheral neuromuscular excitability, the authors speculated that the ergogenic effect of paracetamol was due to a reduction in the magnitude of muscle afferent feedback and consequently an attenuation of neuromuscular fatigue development.

Overall, the human studies on acute exercise performance to date are certainly intriguing (Figure 2), suggesting that analgesic drugs, in this case, paracetamol, can improve both short- and long-term performance parameters through various physiological mechanisms related to the analgesic or antipyretic effect of these drugs. Although the conflicting findings between the early aspirin studies and the later paracetamol studies warrant further research, they indicate potential differences in the biological impact of the drugs on processes related to exercise capacity. Alternatively, differences might simply be due to the disparity in drug dosages or specific protocols of the performance tests used. More importantly, however, these studies have generally been conducted with recreationally trained individuals, and therefore, it remains to be seen whether highly trained elite athletes, already accustomed to training at the highest level (volume and intensity), also experience augmented performance when taking these drugs for competitive advantage. Moreover, because all of these studies are cross-sectional in nature, it is unknown how these interventions might work if used in a chronic training paradigm. Specifically, one might pose the interesting, yet ethically dubious question, whether regular use of analgesics could promote greater tolerance to training stress and hence promote augmented adaptations. Although this sounds like an attractive hypothesis, it will be evident in the next sections that chronically blocking prostaglandin formation might come with negative consequences for the adaptive response.

3 | EXERCISE-INDUCED PROTEIN SYNTHESIS

The first human studies exploring the effects of analgesic drugs on acute muscle adaptive responses came from Todd Trappe's laboratory at Ball State University.[27] Twenty-four young men were assigned to one of three groups that received either ibuprofen (1200 mg/d), paracetamol (4000 mg/d), or a placebo after high-intensity resistance exercise consisting of 10-14 sets of 10 eccentric repetitions with the knee extensors (120% of concentric 1 RM).[27] Postexercise (24 hours) skeletal muscle protein synthesis rates increased 76% in placebo, yet were unchanged in the ibuprofen and the paracetamol group. This increase in protein synthesis was accompanied by increased levels of prostaglandin F2alpha ($PGF_{2\alpha}$) in the placebo group, but not in the two intervention groups.[28] These results suggested that over-the-counter doses of both ibuprofen and paracetamol suppress the protein synthesis response in skeletal muscle after high-intensity muscle-damaging eccentric exercise, and this response was mediated through a common mechanism; that is, reduced $PGF_{2\alpha}$ formation. This is in line with animal data that convincingly showed that NSAIDs attenuate protein synthesis through prostaglandin inhibition.[29,30] In a follow-up study from the Trappe laboratory, the hypothesis was tested that the prostaglandin-mediated increase in protein synthesis was regulated specifically by the COX-2 enzyme.[31] Thus, 16 young men were randomly assigned to either 600 mg of a selective COX-2 inhibitor (celecoxib) or a placebo. Interestingly, postexercise (10 sets of 10 eccentric repetitions at 120% of concentric 1 RM) muscle protein synthesis was not suppressed by the COX-2 inhibitor, suggesting that the COX-1 enzyme could be responsible for the COX-mediated increase in muscle protein synthesis following resistance exercise.

Although these findings suggested that nonselective COX-inhibiting drugs could attenuate muscle protein synthesis, the body of evidence is equivocal. Mikkelsen et al[32] had 8 healthy men complete 200 maximal eccentric contractions with each leg. To block prostaglandin synthesis locally in the skeletal muscle, indomethacin was infused via microdialysis catheters in to *m. vastus lateralis* of one leg. The results showed that myofibrillar and collagen protein synthesis were unaffected by the local NSAID infusion. In addition, in older osteoarthritic patients, ibuprofen did not influence skeletal muscle protein synthesis 24 hours after aerobic exercise.[33] Similar results were noted in elderly men with elevated systemic inflammation, where no effect of a high dose of ibuprofen on the postexercise (3 hours) muscle protein synthetic response following acute resistance exercise was evident.[34] The reason for the discrepancies between studies examining NSAIDs and protein synthesis may be due to the different protocols and measurement techniques. It is possible that the duration of the NSAID infusion (7.5 hours) in the Mikkelsen et al[32] study was not comparable to the orally consumed doses in Trappe's original study.[27] Moreover, changes seen in mixed-muscle protein synthesis[27] could possibly be masked when measuring specific protein fractions, that is, myofibrillar, sarcoplasmic, and collagen protein synthesis.[32,33]

4 | MOLECULAR RESPONSES REGULATING PROTEIN TURNOVER

4.1 | Satellite cell activity

A few studies have assessed the impact of NSAIDs on the myogenic stem cell response to acute exercise bouts. The myogenic stem cell niche, called satellite cells, is activated in

response to exercise and is thought to be important contributors to the repair and remodeling of existing muscle fibers through the formation of new myonuclei.[35,36] Mackey et al[37] explored satellite cell counts in male endurance athletes after a 36-km run. Compared with pre-exercise levels, a 27% increase in the number of satellite cells was observed on day 8 after exercise in the placebo group, while satellite cell levels remained similar in the NSAID group that received 100 mg of indomethacin per day. These results were the first to suggest that ingestion of anti-inflammatory drugs could attenuate the exercise-induced increase in satellite cell number in trained athletes. In a subsequent study,[38] 8 men performed 200 maximal eccentric contractions with each leg, and indomethacin was infused via a microdialysis catheter into the vastus lateralis muscle of one leg (same study as mentioned earlier investigating the protein synthetic response[32]). The main finding was that the NSAID infusion suppressed the exercise-induced increase in the number of satellite cells 8 days after exercise. These results gave further support to the role of COX activity in regulating satellite cell activity after exercise. Interestingly, however, when a COX-2-specific inhibitor (celecoxib) was used in 33 young men and women, neither intramuscular prostaglandin E2 (PGE_2) levels nor satellite cell activity after resistance exercise was altered,[39] indicating that the satellite cell response, just like the protein synthetic response, is regulated by COX-1 rather than COX-2. Contrary to the idea that NSAIDs might dampen the regenerative process, Mackey et al[40] recently reported that satellite cell activation was expedited by 1200 mg of ibuprofen taken 2 weeks before and 4 weeks after an electrical stimulation-induced injury to the leg extensor muscles. Thus, it appears that the nature of the specific exercise and/or damaging challenge probably determines how, and to what extent, satellite cell processes are affected by NSAID administration. Collectively, however, it appears NSAIDs have the capacity to interfere with the normal myogenic stem cell response to acute exercise bouts.

4.2 | Translational signaling

The acute exercise-induced increase in muscle protein synthesis is generally thought to be driven by augmented protein translation, regulated by the mechanistic Target of Rapamycin (mTOR) complex.[41] Two recent studies have explored the acute muscle translational signaling response with or without an analgesic drug in young individuals. In a study performed by Markworth et al[42], 16 healthy male volunteers ingested 1200 mg of ibuprofen (or placebo) in three doses administered both before and following a bout of unaccustomed resistance exercise (3 sets of 8-10 reps at 80% of 1 RM in three different leg exercises). The ibuprofen treatment prevented the sustained elevation of MEK-ERK signaling at 3 hours and 24 hours postexercise, and this was associated with suppressed phosphorylation of ribosomal protein S6. These data suggested that the early translational signaling response could be attenuated with NSAIDs, perhaps explaining the previous reports of an attenuated protein synthetic response after acute resistance exercise.

Recently, similar findings were observed for paracetamol in a double-blind, randomized, crossover study. Eight young men performed two trials of unilateral knee extension resistance exercise (8 sets, 10 reps, 65% of 1 RM) with consumption of either paracetamol (1000 mg/6 hours) or placebo prior to and immediately after the bout.[43] Muscle biopsies were collected at rest and 1 hour and 3 hours postexercise. At 1 hours postexercise, phosphorylation of ribosomal protein S6 was increased in both groups, but to a greater extent in the placebo group. At 3 hours, the phosphorylation of p70S6 kinase was elevated only in placebo. Localization of mTOR to the lysosome (LAMP2) in myosin heavy chain-II fibers increased 3 hours postexercise only in the placebo. Furthermore, myostatin mRNA expression was reduced 1 hour postexercise only in the placebo condition, and myogenic factor 6 (MYF6) mRNA was increased 1 hour and 3 hours postexercise only with paracetamol. Collectively, these studies suggest that both ibuprofen and paracetamol have the potential to modulate early signaling responses that regulate muscle protein turnover.

4.3 | Lipid mediator response

Recent research has highlighted that tissue inflammation and regeneration do not solely work through the COX enzymes.[44] In fact, there are many other lipid mediators with autocrine/paracrine signaling functions that can affect the skeletal muscle in response to exercise challenges. These bioactive lipid mediators, synthesized endogenously from polyunsaturated fatty acids, are mostly known for their key role in the inflammatory response through the classical eicosanoids such as the prostaglandins and leukotrienes. However, classes of lipid mediators with anti-inflammatory and resolving bioactivity, such as lipoxins, resolvins, and protectins, have also been implicated.[44] These mediators act to antagonize the pro-inflammatory response while at the same time actively promoting tissue healing and regeneration.[44] Recent research suggests that these lipid metabolites can have direct regulatory effects on the skeletal muscle. For example, 12/15-hydroxyeicosatetraenoic acids (HETEs) have been found to increase rates of protein breakdown in C2C12 myoblasts and myotubes,[45,46] and several pro-resolving lipid mediators play important roles in satellite cell differentiation and myogenesis.[44,47] Interestingly, a recent study using LC-MS-based lipidomics revealed suppression of both early pro-inflammatory and later anti-inflammatory circulating lipid mediator responses in subjects orally receiving 400 mg ibuprofen.[48] Although this response has not yet been explored

in skeletal muscle, this research opens up the possibility that NSAIDs might indirectly interfere with exercise-induced adaptations by delaying or preventing timely resolution of the inflammatory response. This is in line with the notion of a "recovery dichotomy" where accelerated recovery with the use of interventions might assist in the resolution of function, but could be at the detriment to longer term adaptation and athletic development. Practitioners should therefore identify the primary goal of the training or competition stimulus and decide if shorter-term recovery is a priority (such as in a congested competition period) over the potential adaptive response. The idea of exercise-induced hormesis is something one should be mindful of to facilitate the optimal outcome that must be based on the individual needs and situation, to balance sufficient stimulus with adequate recovery to optimize performance.[49]

In summary, it seems clear that analgesic drugs have the potential to modulate adaptive cellular responses to acute exercise bouts as reflected in the findings on muscle protein synthesis, satellite cell activity, translational signaling, and lipid mediator responses (Figure 2). However, the findings are equivocal, most likely due to the different exercise protocols, subject cohorts (age and training status), and drug types, doses and frequency used. It is important to highlight that much of the experimental work on acute adaptive responses have been done using a very high number of eccentric muscle contractions, which is a powerful muscle-damaging insult that provokes greater muscle stress than most athletes typically would encounter during training sessions. Thus, an important mission for future research is to better characterize the effect of different training variables (eg, exercise mode, muscle contraction type, intensity, volume), as well as different drug treatment (eg, drug type, dose, frequency, and length of treatment) on the acute signaling and subsequent adaptive responses that translate to changes in behavior/performance.

5 | TRAINING ADAPTATIONS

Although analgesic drugs clearly have the capacity to affect acute cellular responses to exercise, this has little relevance for the athlete unless the effects are also translated into altered long-term training adaptations. While animal studies have reported reduced muscle hypertrophy in response to overload with concurrent NSAID administration,[50,51] relatively few studies have assessed whether chronic consumption of analgesic drugs affects training adaptations in humans. Three studies[52-54] explored the effect in older individuals (60 to 80 years old) and one study examined 50- to 70-year-old knee osteoarthritis patients.[55] Although it is questionable whether any inferences to training adaptations in athletes can be made from these studies, it is worth noting that Trappe et al[52] reported that daily doses of 1200 mg ibuprofen or 4000 mg paracetamol during 12 weeks of resistance training resulted in enhanced muscle mass and strength gains of 25%-50% above the placebo-consuming group. These data when coupled with subsequent mechanistic work from the same group,[56] and the cross-sectional data on lean muscle mass and self-reported NSAID use,[57] and the treatment of ibuprofen to limit sarcopenia in older rats,[58] all lend support to the idea that NSAIDs might positively influence training adaptations in older individuals, perhaps related to their higher basal inflammatory state.[59]

Only two studies have assessed the influence of analgesic drugs on specific training adaptations in young individuals (Figure 2). The first study showed that 400 mg of ibuprofen taken immediately after each training session did not influence muscle hypertrophy, strength improvements, or perceived muscle soreness in young (n = 18) resistance-trained subjects.[60] The training protocol consisted of 6 sets of biceps curls 2-3 days/wk for 6 weeks. The drugs were consumed on training days only, resulting in a total intake of 800-1200 mg/wk. Although one could argue that the single-dose administration in close proximity to the training sessions might actually reflect what many athletes would do in real life, the lack of an effect on training adaptations in this study is likely due to the lower total dose taken when compared with previous acute studies where the standard dose has been the maximal-over-the counter dose of 1200 mg ibuprofen (3 × 400 mg over a day).

The second study was a single-blind, randomized controlled training study in young moderately active men and women.[61] The subjects were randomized to either an experimental group receiving 1200 mg ibuprofen (n = 15) or to a control group receiving a very low dose of 75 mg acetylsalicylic acid (n = 16) per day during 8 weeks of resistance exercise of the knee extensor muscles. The results showed that quadriceps muscle volume was significantly attenuated in the ibuprofen group (3.7%) compared with the control group that received low dose aspirin (7.5%). The negative effects of higher doses of ibuprofen were also evident on some, but not all strength measurements. Collectively, these results showed that higher doses of NSAIDs have the capacity to attenuate adaptations induced by resistance training in younger individuals.

The effects of NSAIDs on mitochondrial biogenesis indices have also been examined.[62] Unexpectedly, we noted decreased mitochondrial phosphorylation in response to the resistance training intervention, with no effect of the high or lower NSAID intake. This decreased mitochondrial function occurred despite increased citrate synthase activity that was independent of the drug intake. These findings are in contrast to previous work in the elderly where 12 weeks of resistance training led to greater increases in muscle size and CS activity with ibuprofen.[63] This further highlights the differences between young and elderly in the biological response of NSAIDs taken concurrently with exercise training.

In summary, because only two studies have examined the impact of analgesic drugs on training adaptations in healthy young participants, there are little data to base firm conclusions. Notwithstanding, it appears that higher but not lower doses of NSAIDs have the capacity to diminish some important training adaptations to resistance training. No study has explored the effects of paracetamol on training adaptations in young individuals, but based on the observed similar effects of NSAIDs on translational processes, a negative response would not be surprising. Hitherto, studies in elite athletes or on adaptations related to endurance-type training are completely absent. Given the very strong interest in the potential blunting of the training-induced adaptive responses with recovery interventions, there is a pressing need to understand the implications of analgesic use in well-trained cohorts engaging in strength, endurance, and concurrent exercise training paradigms.

6 | SAFETY AND ETHICAL CONSIDERATIONS

The use of pharmacological methods, in this case using analgesic drugs to modulate exercise performance or adaptations, is an extremely emotive topic and raises ethical concerns and questions about safety, potential side-effects, and issues around drug use for competitive advantage which are tantamount to doping. It also raises questions from a practical point of view. It is easy to realize a moral conflict between using the drugs to be able to compete and train, as opposed to not being able to train or compete at all. Anecdotally, many marathon runners will consume analgesics before and during the race in the belief that there will be some benefit in easing the consequences of the race. However, a relevantly recent study reported the use of analgesics prior to the race did not offer protection from pain during or after the marathon compared with controls.[10] Similarly, the pain sensation associated with muscle damage from eccentric contractions is typically not modulated by these drugs.[12] From an applied perspective, the practitioner and athlete must consider the cost-benefit. On balance, the benefits of taking analgesics to reduce exercise-induced pain look to be very limited indeed, and the risk of potential unwanted side-effects appears to be high, particularly when the doses are higher and prolonged. Even though present data indicate an attenuated training response, support exists at least for paracetamol to potentially enhance performance capacity. Consequently, there are reports strongly advocating that paracetamol should be added to the WADA doping list.[64] This is, however, a complicated matter since there are a plethora of other drugs and substances that might have analgesic effects and to draw a line that defines what substances should be banned and what should be allowed is complex in the extreme, especially since there could be a genuine medical reason for taking them.

Regardless of the doping issue, it seems reasonable to propose that athletes using medicines to reduce pain sensations are likely to be at greater risk of injury and tissue damage. Although the effects of the temporal pattern of NSAIDs use on muscle repair is not known in detail,[15] there are some data suggesting that any potential benefit of NSAIDs in the early phases (first weeks) after extensive fiber damage[40] might not be maintained in the longer term or after more normal physiological loading is possible.[14] Furthermore, chronically dampening the pain response might also lead to athletes returning to play/competition prematurely, thereby increasing the propensity for injury or reoccurrence of previous injuries.

It is also clear that all analgesic drugs come with potential adverse effects. Paracetamol has been associated with liver toxicity, and indeed, paracetamol overdosing is the most common cause of liver failure in the UK.[65] The use of NSAIDs can cause gastrointestinal complications and bleeding.[66] As NSAIDs also affect thrombocyte function and aggregation, there is a risk of more severe bleeding following trauma. This is of genuine concern for athletes in collision sports or where the likelihood of contact is high and hence an increased risk of acute trauma or soft tissue injury. It is important to highlight that athletes taking NSAIDs during activities such as the marathon are reported to have a five times greater incidence of adverse serious events such as gastrointestinal cramps and bleeding, hematuria, cardiovascular events, or temporary kidney failure, which tend to increase with larger analgesic doses.[10] Furthermore, in a large cohort of Finnish Olympic athletes, every fifth individual reported some NSAID-related adverse effect.[67] Moreover, NSAIDs can increase the overall risk of myocardial infarctions, sudden cardiac arrests, and overall mortality rates.[66,68,69] The risks also vary across different analgesics; for example, the increased mortality risk appears to be much higher with diclofenac compared with aspirin or paracetamol.[66] Even though the absolute risk is still very low in healthy individuals, these risks are greater with larger doses and/or with chronic use of these pharmacological interventions.[70]

7 | LIMITATIONS AND DIRECTIONS FOR FUTURE RESEARCH

It is somewhat frustrating that despite the number of studies conducted to date, there are conflicting results and several unresolved questions that are pertinent to the understanding of how analgesic and anti-inflammatory drugs can affect exercise performance and training adaptations. Thus, to date, there is no clear message to the athletes on how they can use these drugs while at the same time minimizing

unwanted effects. Although there are some emerging data suggesting that ingestion of paracetamol in association with isolated exercise performance events might improve performance, it is unknown if this is the case for the highly trained athlete. Thus, there is a need for studies examining the impact of analgesics on applied sport performance outcomes in high-caliber athletes to determine the application to athletic populations. It is also important to acknowledge that we have relatively little information on the precise mechanisms of how analgesics or anti-inflammatories modulate the adaptive training response. As research has clearly shown, the cellular response to acute exercise does not always reflect the end-point adaptations to training and there is an urgent need for studies examining the relationship between the acute adaptive signal and its relationship to the long-term training effect. As evident in this review, the dose and length of drug treatment matters, and to date, we have very little information on any potential dose-response relationships between drug intake and performance and/or adaptations.

8 | PERSPECTIVES

Given the alarmingly high intake of analgesic and anti-inflammatory drugs in elite sports, we encourage greater awareness among athletes, coaches, support staff, and sports organizations about the possible adverse health effects of these drugs. It is also important to highlight that frequent use of NSAIDs, and plausibly also paracetamol, could hamper chronic training adaptations and consequently reduce long-term athletic development. This issue is particularly important to raise, since there is now some preliminary evidence suggesting that these drugs might actually enhance exercise performance in some situations. Thus, it is imperative that we continue to investigate the effects of analgesic drugs on elite athletic performance and training adaptations and that the ethical issues associated with such practices within sports are openly and transparently discussed. Given that over-the-counter painkillers are effective when used as medicinally intended (ie, for alleviating short-term pain symptoms), there is still probably a time and a place for these interventions to be used by elite athletes. Collectively, however, all athletes should, together with coaches, medicine, and science support staff, do a careful risk-benefit analysis before using analgesic and anti-inflammatory drugs regularly in association with training or competition.

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